Anisotropic dielectric function

The dielectric function relates the displacement field to the electric field,

$$\mathbf{D} = \varepsilon_0(1+\chi)\mathbf{E} = \varepsilon_0\varepsilon\mathbf{E}$$

In the static limit, we have $\nabla \times \mathbf{E} = 0$ but, crucially, in an anisotropic material, we do not have $\nabla \times \mathbf{D} = 0$, which is simply because $\nabla \times \mathbf{P} = \nabla \times (\chi\mathbf{E}) \neq 0$, in general. We can, however, write:

$$\mathbf{q}\cdot\mathbf{D} = \varepsilon_0\mathbf{q}\cdot(\varepsilon\mathbf{E})$$

We may also write $i\mathbf{q}\cdot\mathbf{D} = \rho$, $|\mathbf{q}|^2\Phi_{\text{ext}} = \rho/\varepsilon_0$

In calculating the longitudinal-longitudinal response, we find a dielectric function $\epsilon(\mathbf{q}) = \Phi_{\text{ext}}(\mathbf{q})/\Phi_{\text{tot}}(\mathbf{q})$

Therefore, we may write:

$$\Phi_{\text{ext}} = \frac{i\mathbf{q}\cdot\mathbf{D}}{\varepsilon_0|\mathbf{q}|^2} = \epsilon\Phi_{\text{tot}} \rightarrow \frac{\mathbf{q}\cdot(\varepsilon\mathbf{q}\Phi_{\text{tot}})}{|\mathbf{q}|^2} = \epsilon\Phi_{\text{tot}}$$

Therefore, we have, finally,

$$\frac{\mathbf{q}\cdot(\varepsilon\mathbf{q})}{|\mathbf{q}|^2} = \epsilon(\mathbf{q})$$

Note that if ε is a scalar, then we have $\varepsilon = \epsilon$.